Constraint Manifolds and the Limits of Quantum Observability

Resolving the 40–50% Entropy Plateau in NISQ Devices

Kevin Monette

Independent Researcher (AI-assisted research)

kevin.monette@research.org

February 9, 2026 (repository version March 2026)

Abstract

In near-term quantum devices with 16–28+ qubits, state tomography consistently yields von Neumann entropy estimates at 40–50% of the theoretical maximum — a plateau widely attributed to a fundamental decoherence limit of current hardware. We show instead that this plateau is a *measurement artifact arising from finite-sample estimator bias*, not a physical decoherence ceiling.

We formalize the distinction through the constraint manifold $\mathcal{C} \subset \mathcal{D}(\mathcal{H})$ — the subspace of physically accessible density operators — and prove that required measurement resources scale as $\nu \propto 2^{n/2}$ via quantum Fisher information analysis. For typical NISQ tomography budgets ($\nu \sim 10^3$ shots) on systems with $n \geq 15$ qubits, finite-sample estimator bias alone accounts for the observed plateau without invoking physical decoherence.

The framework connects naturally to thermodynamic gravity: constraint-induced entropy gradients source effective stress-energy via the coupling derived in a companion paper [**?**].

Falsification criterion. If entropy estimates for n-qubit coherent states fail to converge to $S_{\mathrm{vN}} < 0.1\,\mathrm{bits}$ when $\nu \geq 100 \cdot 2^{n/2}$ after SPAM calibration, the measurement-insufficiency hypothesis is falsified.

Contents

1 The Constraint Manifold Formalism

1.1 Physically Accessible State Space

Not all density operators in $\mathcal{D}(\mathcal{H})$ are physically accessible to a quantum system evolving under real conditions. Conservation laws, decoherence channels, and initial-state constraints

restrict evolution to a proper sub-manifold.

Definition 1.1 (Constraint Manifold)**.**

$$\mathcal{C} \;=\; \big\{\, \rho \in \mathcal{D}(\mathcal{H}) \;\big|\; \mathrm{Tr}\big(\hat{C}_i\, \rho\big) = c_i \;\; \forall\, i \in \mathcal{I}_{\mathrm{irr}} \big\}, \tag{1}$$

where $\hat{C}_i$ are constraint operators (e.g. $\hat{H}$ for energy conservation, $\hat{Q}$ for charge), c_i are the constraint values fixed by initial conditions, and $\mathcal{I}_{\mathrm{irr}}$ indexes *irreversible* constraints — those that cannot be undone by subsequent unitary evolution.

Soft constraints (e.g. thermodynamic bias toward equilibrium) enter through a measure μ on $\mathcal{C}$ rather than its definition:

$$\mu(d\rho) \;\propto\; e^{-\beta\, \mathrm{Tr}(\hat{H}\rho)}\, d\rho. \tag{2}$$

1.2 The Observable Subspace

A measurement history $r = \{i_1, i_2, \ldots, i_k\}$ represents the sequence of constraint-fixing events that have become irreversible. Each event reduces the accessible subspace:

$$\mathcal{O}(r) \;=\; \big\{\, \rho \in \mathcal{C} \;\big|\; \mathrm{Tr}\big(\hat{C}_{i_j}\, \rho\big) = c_{i_j} \;\; \forall\, j \le k \big\}. \tag{3}$$

As k increases, $\mathcal{O}(r)$ shrinks. This is not decoherence — it is the progressive fixing of constraints by observation.

Proposition 1.2 (Dimension of the Observable Subspace)**.** *For an n-qubit system with m linearly independent measurement constraints,*

$$\dim \mathcal{O}(r) \;=\; 4^n - \mathrm{rank}(\mathcal{C}) - 1, \tag{4}$$

where $\mathrm{rank}(\mathcal{C})$ *is the number of linearly independent constraints.*

Proof. The space of n-qubit density operators (Hermitian, positive semidefinite, trace-1) has real dimension $4^n - 1$ (the $2^n \times 2^n$ Hermitian matrix parameters minus the trace normalization). Each linearly independent irreversible constraint reduces the dimension by one. Linear independence must be verified in the space of Hermitian operators; redundant constraints reduce the effective rank below m. □

Remark 1.3. When $m \ll 4^n$, the observable subspace vastly undersamples the physical state space, creating an apparent entropy increase without any physical decoherence.

2 Quantum Fisher Information and Shot Scaling

2.1 The Quantum Cramér–Rao Bound

The quantum Cramér–Rao bound places a fundamental lower limit on the variance of any unbiased estimator $\hat{S}_{\mathrm{vN}}$ of the von Neumann entropy:

$$\mathrm{Var}[\hat{S}_{\mathrm{vN}}] \;\ge\; \frac{\big[F_Q(\rho)\big]^{-1}}{\nu}, \tag{5}$$

where ν is the number of independent measurement shots and $F_Q(\rho) = \mathrm{Tr}[\rho\, L^2]$ is the quantum Fisher information, with L the symmetric logarithmic derivative satisfying $\partial_\theta \rho = (\rho L + L\rho)/2$.

2.2 Exponential Scaling Near Maximum Entropy

Lemma 2.1 (Fisher Information Scaling)**.** *For states near the maximally mixed state* $\rho \approx \mathbf{I}/2^n$*,*

$$F_Q(\rho) \;\sim\; 2^{-n/2}. \tag{6}$$

Near maximum entropy, all probability distributions over measurement outcomes are nearly identical. Distinguishing $S_{\mathrm{vN}} = 0.99\, S_{\max}$ from $S_{\mathrm{vN}} = S_{\max}$ requires resolving tiny differences in an exponentially large space of nearly equally likely outcomes. The sensitivity of the measurement record to entropy changes scales as the average overlap between adjacent density matrices in the entropy-variation direction, which is exponentially close to unity in any product measurement basis.

2.3 Required Shot Count

To achieve precision ε in the entropy estimate, the Cramér–Rao bound requires:

$$\nu \;\gtrsim\; \varepsilon^{-2} \cdot 2^{n/2}. \tag{7}$$

Table 1: Required measurement shots for entropy estimation at precision $\varepsilon = 0.1$ bits.

Qubit count n	Hilbert dim. $d = 2^n$	Required ν	Typical NISQ ν	Gap factor
5	32	≈560	10^3	<1 (sufficient)
10	1 024	≈3 160	10^3	≈3×
15	32 768	≈17 800	10^3	≈18×
20	1 048 576	≈100 000	10^3	≈100×
25	≈ 33M	≈560 000	10^3	≈560×
28	≈ 268M	≈1.6M	10^3	≈1 600×

The gap between required and typical shot budgets explains the entropy plateau for $n \geq 15$ entirely: NISQ devices are being measured with 18–1 600× too few shots to accurately estimate entropy.

3 The Estimator Bias Formula

3.1 Linear Inversion Bias

The standard linear-inversion tomographic estimator exhibits systematic bias scaling with Hilbert space dimension $d = 2^n$ [**?**]:

$$\mathbb{E}[\hat{S}_{\mathrm{vN}}] \;=\; S_{\mathrm{vN}}(\rho) \;+\; \underbrace{\frac{d-1}{2\nu}}_{\text{finite-sampling bias}} \;+\; \underbrace{B_{\mathrm{SPAM}}}_{\text{readout errors}} \;+ O\big(\nu^{-2}\big)\,. \tag{8}$$

The estimator *systematically overestimates* entropy by $(d-1)/(2\nu)$ bits, independent of the true entropy and regardless of hardware quality.

3.2 Quantitative Explanation of the 40–50% Plateau

Example: $n = 15$ **qubits,** $\nu = 10^4$ **shots.**

$$\begin{aligned} d &= 2^{15} = 32\,768, \\ \text{Finite-sampling bias} &= \frac{d-1}{2\nu} = \frac{32\,767}{20\,000} \approx 1.64 \text{ bits}, \\ S_{\max} &= n \ln 2 \approx 10.4 \text{ bits}, \\ \text{Bias as fraction of } S_{\max} &\approx 15.8\%. \end{aligned} \tag{9}$$

Adding SPAM errors ($B_{\text{SPAM}} \approx 0.5$–$1.0$ bits for current hardware):

$$\text{Total apparent inflation} \approx \frac{1.64 + 0.75}{10.4} \approx 23\%. \tag{10}$$

For a state with true $S_{\text{vN}} \approx 20$–25% of $S_{\max}$, the measured value lands at ≈ 43–48% — precisely matching the observed NISQ plateau without physical decoherence.

3.3 The Bias-Corrected Estimator

Define the corrected estimator:

$$\hat{S}_{\text{vN}}^{\text{corr}} \;=\; \hat{S}_{\text{vN}} - \frac{d-1}{2\nu} - \hat{B}_{\text{SPAM}}, \tag{11}$$

where $\hat{B}_{\text{SPAM}}$ is estimated from measurement calibration circuits.

Under the measurement-insufficiency hypothesis, the corrected estimator satisfies:

$$\hat{S}_{\text{vN}}^{\text{corr}}(\nu) \;=\; S_{\text{vN}}^{\text{phys}}(\rho) \;+\; \frac{d-1}{2\nu} \;+\; O\big(\nu^{-2}\big)\,, \tag{12}$$

converging monotonically to the true physical entropy as $\nu \to \infty$.

4 Connection to Thermodynamic Gravity

The constraint manifold formalism connects naturally to the entropic gravity program. In Jacobson's thermodynamic derivation [**?**], spacetime geometry emerges from entropy gradients across causal horizons. The companion paper [**?**] extends this to laboratory scales.

Constraint-induced entropy gradients ∇S_{vN} source effective stress-energy via:

$$\text{(entropy gradient)} \;\longrightarrow\; \text{effective stress-energy} \;\longrightarrow\; \text{spacetime curvature.} \tag{13}$$

For the gravitational prediction to be testable, experiments must measure *physical* entropy changes, not artifacts of the measurement budget. Accordingly, the shot-scaling requirement (**??**) is a prerequisite for any entanglement-gravity experiment: SPAM-corrected, high-ν tomography must be performed before interpreting any entropy difference as a gravitational source.

Observation is not passive. An observation is any physical process that irreversibly correlates with a constraint value of the observed system. Decoherence is environmental observation. Conscious measurement is the same physical process, more controlled. Consciousness plays no special role — the constraint-fixing is thermodynamic, not mental.

5 Experimental Protocol for Falsification

5.1 Three-Stage Validation Protocol

Stage 1 — Calibration. Prepare known-entropy states (computational basis states, Bell pairs, product states) with theoretically exact S_{vN}. Construct the SPAM correction matrix Λ from calibration circuits. Verify that $\hat{S}_{\mathrm{vN}}^{\mathrm{corr}}$ converges to theoretical S_{vN} at rate $O(1/\nu)$.

Stage 2 — Scaling test. Prepare n-qubit GHZ states $|\mathrm{GHZ}\rangle = (|0\rangle^{\otimes n} + |1\rangle^{\otimes n})/\sqrt{2}$ for $n \in \{5, 8, 10, 12, 15\}$. Measure $\hat{S}_{\mathrm{vN}}(\nu)$ for $\nu \in \{10^3, 10^4, 10^5, 10^6\}$ shots. Apply SPAM correction: $\hat{\rho}_{\mathrm{corr}} = \Lambda^{-1}\hat{\rho}_{\mathrm{raw}}$.

Stage 3 — Convergence verification. Compute $\hat{S}_{\mathrm{vN}}^{\mathrm{corr}}(\nu)$ from (**??**). Fit to the scaling model:

$$\hat{S}_{\mathrm{vN}}^{\mathrm{corr}}(\nu) = S_{\mathrm{vN}}^{\mathrm{phys}} + \frac{A}{\nu} + O\big(\nu^{-2}\big)\,. \tag{14}$$

Extract $S_{\mathrm{vN}}^{\mathrm{phys}}$ as the $\nu \to \infty$ extrapolation and compare to the theoretical prediction ($S_{\mathrm{vN}} = 0$ for pure GHZ).

5.2 Falsification Criterion

Falsification: The measurement-insufficiency hypothesis is falsified if, for n-qubit GHZ states with verified entanglement depth $\geq n$, the SPAM-corrected entropy estimate $\hat{S}_{\mathrm{vN}}^{\mathrm{corr}}$ fails to converge to $S_{\mathrm{vN}} < 0.1$ bits when shot count $\nu \geq 100 \cdot 2^{n/2}$, verified by bootstrap resampling with $> 95\%$ confidence.

6 Implications

6.1 For Quantum Computing

Many NISQ devices have better coherence properties than their entropy estimates suggest. Quantum-advantage assessments based on entropy metrics should account for the estimator bias of Eq. (**??**) before concluding that hardware is at a decoherence limit.

6.2 For Quantum Gravity Experiments

Physical entropy measurement (not estimated entropy) is a prerequisite for any entanglement-gravity experiment. The shot-scaling requirement (**??**) must be satisfied before any differential entropy is attributed to a gravitational source.

6.3 For Quantum Information

The scaling law $\nu \propto 2^{n/2}$ is a fundamental resource bound for entropy estimation in n-qubit systems. This is sub-exponential compared to full state tomography ($\nu \propto 4^n$) and can in principle be achieved by randomized measurement protocols [**?**].

7 Conclusion

The apparent entropy plateau at 40–50% in NISQ devices is a consequence of finite-sample estimator bias, not fundamental hardware limitations. Required shot count for accurate entropy estimation scales as $\nu \propto 2^{n/2}$, derived rigorously from quantum Fisher information analysis. The measurement-insufficiency hypothesis makes precise, falsifiable predictions verifiable with existing hardware.

The framework establishes three results for quantum physics broadly:

1. Physical entropy and measured entropy are distinct quantities, separated by a computable, correctable bias.

2. Observation is a physical resource (measurement shots) that must scale with Hilbert space dimension.

3. Much apparent disorder in NISQ systems is epistemic (measurement-limited), not physical (decoherence-limited).

Acknowledgments

The author acknowledges AI research partners for assistance with literature synthesis, equation formatting, and critical review. All theoretical content and experimental design originate from the author's independent research program.

References

[1] T. Jacobson, "Thermodynamics of spacetime: The Einstein equation of state," *Phys. Rev. Lett.* **75**, 1260 (1995). doi:10.1103/PhysRevLett.75.1260

[2] K. Monette, "Gravitational coupling to entanglement entropy density," Independent Research Program, March 2026.

[3] S. T. Flammia, D. Gross, Y.-K. Liu, and J. Eisert, "Quantum tomography via compressed sensing: error bounds, sample complexity, and efficient estimators," *New J. Phys.* **14**, 095022 (2012). doi:10.1088/1367-2630/14/9/095022

[4] M. G. A. Paris, "Quantum estimation for quantum technology," *Int. J. Quantum Inf.* **7**, 125 (2009). doi:10.1142/S0219749909004839

[5] J. Cotler and F. Wilczek, "Quantum overlapping tomography," *Phys. Rev. Lett.* **124**, 100401 (2020). doi:10.1103/PhysRevLett.124.100401

[6] E. Knill *et al.*, "Randomized benchmarking of quantum gates," *Phys. Rev. A* **77**, 012307 (2008).